=== UNIT 6: KINEMATICS AND KINETICS OF RIGID BODIES ===

>> 6.1 Planar Kinematics: Translation, Fixed Rotation, and General Motion

A rigid body in planar motion experiences three distinct kinematic modes:
* 1. Pure Translation: Every straight line in the body remains strictly parallel to its original direction. All points possess identical velocity and acceleration vectors at any instant: v_A = v_B, and a_A = a_B.
* 2. Fixed-Axis Rotation: All particles move in concentric circular paths about a stationary fixed axis perpendicular to the plane of motion. The kinematics are defined by angular displacement theta(t), angular velocity omega = dtheta/dt, and angular acceleration alpha = domega/dt.
- Linear Velocity: v = r * omega (directed tangentially).
- Tangential Acceleration: a_t = r * alpha.
- Normal (Centripetal) Acceleration: a_n = r * omega^2 = v^2 / r.
- Total Linear Acceleration: a = sqrt(a_t^2 + a_n^2) = r * sqrt(alpha^2 + omega^4).
* 3. General Plane Motion: A simultaneous combination of translation and rotation: v_B = v_A + v_B/A, where v_B/A = omega x r_B/A.

>> 6.2 Instantaneous Centre of Zero Velocity (ICR / ICZ)

At any given instant, the general planar motion of a rigid body can be conceptualized as a pure rotation about a single point in the plane having ZERO linear velocity, termed the Instantaneous Centre of Rotation (ICR):
* Locating the ICR:
- Case 1: If velocities v_A and v_B at two distinct points A and B are known in direction (non-parallel), the ICR is the geometric intersection of perpendicular lines drawn to v_A and v_B.
- Case 2: For a cylinder or wheel rolling WITHOUT SLIPPING on a stationary surface, the contact point is instantaneously at rest. Hence, the contact point is the ICR.
* Velocity Distribution from ICR: Once ICR is found, the velocity of ANY point P on the body is: v_P = omega * r_P/ICR, directed perpendicular to line P-ICR.

>> 6.3 Kinetics of Rigid Bodies: Equations of Motion

For a rigid body of mass m with centre of mass G subjected to general coplanar force systems:
* Translational Equations of Motion: Sigma Fx = m * a_Gx, and Sigma Fy = m * a_Gy.
* Rotational Equation of Motion: Sigma M_G = I_G * alpha, where I_G is the mass moment of inertia about the centroidal z-axis.
* Alternative Moment Equation about arbitrary point P: Sigma M_P = I_G * alpha + (m * a_G) * d.

* Fixed-Axis Rotation about fixed hinge pin O: Sigma M_O = I_O * alpha, where I_O = I_G + m*d^2 (by parallel axis theorem).

>> 6.4 Work-Energy Principle for Rigid Bodies in Planar Motion

* Kinetic Energy of a Planar Rigid Body: KE = 0.5 * m * (v_G)^2 + 0.5 * I_G * (omega)^2.
- Pure Translation (omega = 0): KE = 0.5 * m * v_G^2.
- Pure Rotation about fixed pin O (v_O = 0): KE = 0.5 * I_O * omega^2.
- Rolling Without Slipping (v_G = R*omega): KE = 0.5 * m * (R*omega)^2 + 0.5 * I_G * omega^2 = 0.5 * I_ICR * omega^2.
* Work-Energy Theorem: Sigma U_1-2 = KE_2 - KE_1.
Work of a Couple/Torque: U = Integral(M dtheta) = M * Delta theta (for constant torque M).

>> 6.5 Angular Momentum and Impulse-Momentum for Rigid Bodies

* Angular Momentum about G: H_G = I_G * omega.
* Angular Momentum about fixed pin O: H_O = I_O * omega.
* Principle of Angular Impulse and Momentum: Integral_t1^t2(Sigma M_G dt) = I_G * omega_2 - I_G * omega_1.
* Conservation of Angular Momentum: If the resultant external moment about an axis is zero (Sigma M_ext = 0), total angular momentum about that axis is conserved: I_1 * omega_1 = I_2 * omega_2.

[EXAMPLE 1: Velocity Analysis of Crank Mechanism via ICR]

Problem Context: A reciprocating slider-crank mechanism has a crank OA of length r = 150 mm rotating counterclockwise at constant speed N = 300 RPM (omega = 31.42 rad/s). Connecting rod AB has length L = 500 mm. At the instant when crank OA makes theta = 60 deg with the horizontal line of slider stroke, calculate: (1) Linear velocity of crank pin A, (2) Angular velocity omega_AB of the connecting rod, (3) Linear velocity of slider B.
Solution Step 1: Linear Velocity of Crank Pin A:
v_A = r * omega = 0.150 m * 31.42 rad/s = 4.71 m/s.
Direction of v_A: Perpendicular to crank OA (at angle 60 + 90 = 150 deg).
Solution Step 2: Establish Geometry and Locate ICR of Rod AB:
Slider B is constrained to move horizontally along the stroke axis => velocity v_B is strictly horizontal.
- Draw perpendicular to v_A at A (which is line OA extended).
- Draw perpendicular to v_B at B (which is a vertical line through B).
- Intersection of line OA extended and vertical line through B defines the ICR.

In triangle O-B-ICR:
Angle of rod AB with horizontal: In triangle OAB, by sine rule: 0.150 / sin(beta) = 0.500 / sin(60) => sin(beta) = 0.150 * 0.8660 / 0.500 = 0.2598 => beta = 15.06 deg.
Distance from ICR to A: r_A/ICR = 0.500 * [cos(15.06) / sin(60)] = 0.500 * (0.9657 / 0.8660) = 0.5576 m.
Distance from ICR to B: r_B/ICR = 0.150 * [sin(60 + 15.06) / sin(60)] + ... = 0.4430 m.
Solution Step 3: Compute Angular Velocity omega_AB and Slider Velocity v_B:
omega_AB = v_A / r_A/ICR = 4.71 m/s / 0.5576 m = 8.45 rad/s (clockwise).
v_B = omega_AB * r_B/ICR = 8.45 rad/s * 0.4430 m = 3.74 m/s.
Conclusion: Crank pin velocity v_A = 4.71 m/s; connecting rod omega_AB = 8.45 rad/s; slider velocity v_B = 3.74 m/s.

[EXAMPLE 2: Cylinder Rolling Without Slipping Down an Incline]

Problem Context: A solid uniform civil drainage concrete pipe section of mass m = 200 kg and outer radius R = 0.40 m (centroidal MOI I_G = 0.5 * m * R^2) is released from rest on an incline of theta = 30 deg. Assuming it rolls without slipping, calculate: (1) Linear acceleration a_G of its center of mass, (2) Minimum coefficient of static friction mu_s required to prevent slipping.
Solution Step 1: Establish Equations of Motion for Rolling Body:
Forces acting: Gravity component down incline m*g*sin(30); Normal reaction N = m*g*cos(30); Friction force F_s acting up the incline at contact point.
Translational equilibrium down incline: m * g * sin(30) - F_s = m * a_G. (Equation 1)
Rotational equilibrium about center G: F_s * R = I_G * alpha. (Equation 2)
Kinematic rolling condition (no slip): a_G = R * alpha => alpha = a_G / R.
Solution Step 2: Substitute rolling condition into rotational equation:
F_s * R = (0.5 * m * R^2) * (a_G / R) = 0.5 * m * R * a_G => F_s = 0.5 * m * a_G.
Solution Step 3: Solve for Linear Acceleration a_G:
Substitute F_s into Equation 1: m * g * sin(30) - 0.5 * m * a_G = m * a_G.
1.5 * m * a_G = m * g * sin(30) => a_G = (2/3) * g * sin(30).
a_G = (2/3) * 9.81 * 0.50 = 3.27 m/s^2.
Solution Step 4: Calculate Friction Force and Minimum Friction Coefficient:
F_s = 0.5 * 200 * 3.27 = 327.0 N.
Normal Reaction: N = m * g * cos(30) = 200 * 9.81 * 0.8660 = 1699.1 N.
Condition for no slip: F_s <= mu_s * N => mu_s >= F_s / N.
mu_s >= 327.0 / 1699.1 = 0.1925 (or mu_s >= tan(30) / 3 = 0.5774 / 3 = 0.1925).
Conclusion: Acceleration a_G = 3.27 m/s^2; Minimum coefficient of static friction mu_s = 0.193.

[EXAMPLE 3: Flywheel Braking Torque and Deceleration]

Problem Context: A heavy cast-iron flywheel of mass m = 500 kg and radius of gyration k = 0.60 m rotates at N1 = 600 RPM. A mechanical friction brake is applied, exerting a constant retarding braking torque of T_brake = 180 N-m. Determine: (1) Initial kinetic energy of the flywheel, (2) Angular deceleration alpha, (3) Time t required to bring the flywheel to rest, (4) Total revolutions N_rev turned during braking.
Solution Step 1: Mass Moment of Inertia I_G and Initial Angular Velocity omega_1:
I_G = m * k^2 = 500 * (0.60)^2 = 500 * 0.36 = 180.0 kg-m^2.
omega_1 = (2 * pi * N1) / 60 = (2 * pi * 600) / 60 = 20 * pi = 62.83 rad/s.
Solution Step 2: Initial Kinetic Energy KE_1:
KE_1 = 0.5 * I_G * (omega_1)^2 = 0.5 * 180.0 * (62.83)^2 = 90.0 * 3947.84 = 355,305.6 J = 355.31 kJ.
Solution Step 3: Angular Deceleration alpha:
T_brake = I_G * alpha => alpha = T_brake / I_G = 180 N-m / 180 kg-m^2 = 1.0 rad/s^2 (retarding).
Solution Step 4: Stopping Time t and Revolutions N_rev:
omega_2 = omega_1 - alpha * t => 0 = 62.83 - 1.0 * t => t = 62.83 s (approx 1 minute 3 seconds).
Total angular displacement: Delta theta = (omega_1)^2 / (2 * alpha) = (62.83)^2 / (2 * 1.0) = 1973.92 radians.
Number of revolutions: N_rev = Delta theta / (2 * pi) = 1973.92 / 6.2832 = 314.16 revolutions.
Conclusion: KE_initial = 355.31 kJ; stopping time t = 62.83 s; revolutions turned = 314.16 rev.

[EXAMPLE 4: Swinging Slender Structural Rod Released from Rest]

Problem Context: A uniform slender steel member AB of mass m = 24 kg and length L = 2.0 m is hinged at end A and held horizontally. It is released from rest. At the instant it swings through the vertical position, calculate: (1) Angular velocity omega, (2) Angular acceleration alpha, (3) Total hinge reaction force at A.
Solution Step 1: Conservation of Energy between horizontal (1) and vertical (2):
Hinge MOI for slender rod: I_A = (1/3) * m * L^2 = (1/3) * 24 * (2.0)^2 = 32.0 kg-m^2.
Vertical drop of center of mass G: Delta h = L / 2 = 2.0 / 2 = 1.0 m.
Potential energy loss = m * g * Delta h = 24 * 9.81 * 1.0 = 235.44 J.
Kinetic energy in vertical position = 0.5 * I_A * omega^2 = 0.5 * 32.0 * omega^2 = 16.0 * omega^2.
Equating: 16.0 * omega^2 = 235.44 => omega^2 = 14.715 => omega = sqrt(14.715) = 3.84 rad/s.
Solution Step 2: Angular Acceleration alpha in vertical position:
In vertical position, line of gravity force passes directly through hinge pin A =>
Moment arm d = 0.
Sigma M_A = m * g * 0 = 0 => alpha = 0 rad/s^2.
Solution Step 3: Reaction Force at Hinge A in vertical position:
Center of mass G experiences centripetal acceleration upward toward hinge A:

a_Gn = (L / 2) * omega^2 = 1.0 m * 14.715 = 14.715 m/s^2.
Dynamic equilibrium vertically (taking upward as positive): R_A - m*g = m * a_Gn.
R_A = m * (g + a_Gn) = 24 * (9.81 + 14.715) = 24 * (24.525) = 588.60 N.
(Notice R_A = 2.5 * m*g = 2.5 * 235.44 = 588.60 N).
Conclusion: At the vertical position: omega = 3.84 rad/s; alpha = 0; vertical hinge reaction R_A = 588.60 N (upward).

>> 6.6 University Review & Examination Questions

* Q1: Define the Instantaneous Centre of Rotation (ICR) and explain how it is used to determine velocity distribution in a planar mechanism.
Answer: The Instantaneous Centre of Rotation is a unique point in the plane of a rigid body experiencing general planar motion that instantaneously possesses zero linear velocity. Since its velocity is zero at that particular instant, the complex plane motion (translation + rotation) of the body reduces to pure rotation about this ICR. To locate the ICR: if the direction of velocity vectors at two distinct points A and B are known, perpendicular lines drawn to these velocity vectors intersect at the ICR. The magnitude of velocity of any arbitrary point P on the body is then simply v_P = omega * r_P/ICR, and its direction is perpendicular to the radial vector connecting ICR to P.
* Q2: Derive the general kinetic energy expression for a rigid body undergoing general planar motion.
Answer: Consider a rigid body composed of particles of mass m_i with position vector r_i relative to center of mass G. Let G have translational velocity v_G and the body rotate with angular velocity omega. The velocity of particle i is v_i = v_G + (omega x r_i). Kinetic energy is: KE = 0.5 * Sigma(m_i * |v_i|^2) = 0.5 * Sigma(m_i * [v_G^2 + 2*v_G*(omega x r_i) + |omega x r_i|^2]). Since Integral r_i dm = 0 (by definition of center of mass), the cross term vanishes identically. The term Sigma(m_i * |omega x r_i|^2) equals I_G * omega^2. Therefore: KE = 0.5 * m * v_G^2 + 0.5 * I_G * omega^2. The total kinetic energy separates neatly into translational energy of the center of mass and rotational energy about the center of mass.

>> 6.7 Key Takeaways and Revision Points

* General planar motion decomposes into translation of center of mass G plus rotation about G.
* At the ICR, velocity is zero; v_P = omega * r_P/ICR for every point on the body.
* Rolling without slipping has ICR at the instantaneous point of ground contact.
* Kinetic energy of rolling body combines translation and rotation: KE = 0.5*m*v_G^2 + 0.5*I_G*omega^2.